Online Resource 1

Calibration-guided robust reallocation of emergency resources under evolving flood risk forecasts: a digital twin framework. D. Mounika and J. Kolluri. *Natural Hazards*.

Contents: S1 experiment configuration; S2 allocation results with interquartile ranges (2018, 2019); S3 sensitivity to α and η; S4 alternative need-share assumptions; S5 run log with input checksums. Code and notebooks: https://github.com/dmounikaresearch-cmyk/kerala-flood-cgrr.

S1 Experiment configuration

Parameter	Value
smoke_test	False
events	['2018', '2019']
model.alpha	0.1
model.eta	0.1
model.beta	0.5
model.tau_max	12.0
model.lam	0.01
model.mu	0.05
model.rho	0.5
model.fixed_width	0.3
model.b4_gamma_frac	0.5
model.need_beta	[2.0, 98.0]
model.resources.team.kappa	150
model.resources.team.hmax	0.3
model.resources.team.mode	road
model.resources.boat.kappa	60
model.resources.boat.hmax	99.0
model.resources.boat.mode	boat
model.resources.medical.kappa	80
model.resources.medical.hmax	0.3
model.resources.medical.mode	road
experiment.n_runs	200
experiment.n_scen_b3	30
experiment.b5_search	60
experiment.seed	42
experiment.sensitivity_alpha	[0.05, 0.1, 0.2, 0.3]
experiment.sensitivity_eta	[0.0, 0.05, 0.1, 0.2, 0.3]
experiment.demand_means	[0.01, 0.02, 0.05]

Values marked ASSUMED in the configuration file: β, $\tau^{\max}$, need share Beta(2, 98), resource capacities κ_r and depth limits $h_r^{\max}$. Stocks: 2018, 55 teams, 110 boats, 16 medical units; 2019, 9 teams, 18 boats, 8 medical units, across the hubs Thrissur, Ernakulam, Thiruvananthapuram and Kozhikode.

S2 Allocation results, 2018 (median [interquartile range] over 200 runs)

Method	Worst-unit weighted unmet	Total unmet (people)	Mean response (h)	P95 response (h)	Coverage (%)
B1	138,870 [89,500, 187,000]	281,293 [198,000, 387,000]	0.78 [0.64, 0.88]	1.48 [1.45, 1.67]	36.3 [28.6, 42.2]
B2	82,750 [54,500, 147,000]	159,526 [99,500, 253,000]	1.11 [0.89, 1.23]	2.75 [2.09, 2.81]	63.0 [53.5, 71.4]
B3	53,543 [15,000, 117,000]	69,147 [22,600, 150,000]	0.96 [0.75, 1.09]	2.08 [2.08, 2.08]	85.2 [73.9, 94.2]
B4	101,830 [60,900, 154,000]	152,695 [91,900, 246,000]	0.78 [0.64, 0.88]	1.48 [1.45, 1.67]	65.6 [53.5, 74.7]
B5	82,481 [54,300, 126,000]	167,604 [109,000, 232,000]	1.13 [0.90, 1.25]	2.75 [2.09, 2.81]	62.7 [57.1, 69.5]
CGRR	72,937 [30,400, 132,000]	101,557 [46,500, 180,000]	0.78 [0.64, 0.89]	1.48 [1.45, 1.67]	76.5 [66.1, 87.4]
FU	116,613 [76,000, 166,000]	192,859 [127,000, 294,000]	0.78 [0.64, 0.88]	1.48 [1.45, 1.67]	55.6 [44.8, 63.8]
FC	114,526 [71,600, 163,000]	182,789 [119,000, 290,000]	0.78 [0.64, 0.88]	1.48 [1.45, 1.67]	57.7 [46.2, 66.4]
A1	101,830 [60,900, 154,000]	153,051 [91,900, 246,000]	0.78 [0.64, 0.88]	1.48 [1.45, 1.67]	65.4 [53.5, 74.7]
A2a	72,972 [30,500, 132,000]	101,592 [46,500, 180,000]	0.78 [0.64, 0.89]	1.48 [1.45, 1.67]	76.5 [66.0, 87.3]
A2b	104,305 [67,300, 153,000]	185,412 [122,000, 281,000]	0.86 [0.71, 0.97]	1.67 [1.45, 1.67]	56.9 [47.1, 66.0]
A3	70,918 [29,300, 133,000]	101,257 [45,400, 179,000]	0.83 [0.69, 0.96]	1.67 [1.45, 1.67]	76.6 [66.1, 87.9]

Method	Missed high-risk	Relocations	Re-plans	Failed dispatches	Gini	Utilization (%)
B1	5.0	0	1	0	0.22	60.7
B2	3.0	997	96	1	0.11	17.6
B3	1.0	190	96	5	0.03	23.9
B4	2.0	77	96	0	0.09	44.1
B5	3.0	5,547	96	1	0.11	17.8
CGRR	1.0	109	20	1	0.06	35.3
FU	3.5	43	96	0	0.13	53.2
FC	3.0	63	96	0	0.11	49.5
A1	2.0	76	21	0	0.09	44.3
A2a	1.0	109	96	1	0.06	35.4
A2b	3.0	0	1	24	0.15	57.1
A3	2.0	75	21	19	0.07	35.3

S2 Allocation results, 2019 (median [interquartile range] over 200 runs)

Method	Worst-unit weighted unmet	Total unmet (people)	Mean response (h)	P95 response (h)	Coverage (%)
B1	123,695 [88,300, 166,000]	293,336 [219,000, 387,000]	0.94 [0.74, 1.12]	2.08 [2.08, 2.08]	29.5 [24.1, 35.5]
B2	122,991 [88,000, 167,000]	301,030 [223,000, 397,000]	1.28 [1.02, 1.46]	3.72 [3.72, 3.72]	27.8 [22.8, 33.7]
B3	125,563 [88,000, 170,000]	299,584 [227,000, 396,000]	1.14 [0.93, 1.30]	2.08 [2.08, 2.08]	27.6 [23.2, 33.4]
B4	125,132 [88,000, 171,000]	299,034 [224,000, 391,000]	2.07 [1.86, 2.25]	12.00 [12.00, 12.00]	28.2 [23.4, 34.0]
B5	122,991 [88,000, 167,000]	301,110 [223,000, 397,000]	1.28 [1.02, 1.47]	3.72 [3.72, 3.72]	27.8 [22.7, 33.7]
CGRR	117,669 [82,600, 157,000]	305,489 [226,000, 392,000]	1.44 [1.16, 1.70]	12.00 [2.08, 12.00]	28.1 [23.0, 33.8]
FU	124,541 [87,200, 169,000]	301,700 [227,000, 396,000]	1.23 [1.03, 1.43]	1.73 [1.73, 2.90]	27.4 [22.6, 32.9]
FC	123,682 [86,200, 168,000]	299,309 [223,000, 395,000]	1.17 [1.02, 1.27]	2.90 [2.73, 2.90]	28.0 [23.3, 33.9]
A1	122,736 [89,000, 166,000]	297,445 [223,000, 392,000]	1.82 [1.64, 2.03]	12.00 [12.00, 12.00]	28.5 [23.7, 34.2]
A2a	119,375 [84,600, 161,000]	305,843 [226,000, 390,000]	1.42 [1.15, 1.67]	3.43 [2.08, 12.00]	28.1 [23.0, 33.7]
A2b	117,518 [86,400, 156,000]	296,928 [219,000, 389,000]	1.10 [0.90, 1.30]	2.08 [2.08, 2.08]	29.5 [23.3, 35.2]
A3	118,358 [86,100, 163,000]	290,451 [215,000, 383,000]	2.64 [1.91, 3.65]	12.00 [12.00, 12.00]	30.8 [24.0, 36.7]

Method	Missed high-risk	Relocations	Re-plans	Failed dispatches	Gini	Utilization (%)
B1	4.0	0	1	0	0.26	48.7
B2	4.0	1,365	84	1	0.31	45.8
B3	4.0	735	84	1	0.30	45.8
B4	4.0	579	84	1	0.35	46.7
B5	4.0	1,364	84	1	0.31	45.7
CGRR	3.0	391	69	0	0.37	45.7
FU	4.0	349	84	1	0.29	45.9
FC	3.0	657	84	1	0.29	46.1
A1	4.0	524	68	1	0.34	47.6
A2a	4.0	466	84	0	0.36	45.8
A2b	3.0	0	1	0	0.46	46.9
A3	3.0	9	64	0	0.40	49.4

S3 Sensitivity to α and η, 2018 (CGRR, median over 200 runs)

α	η	Set coverage	Mean set size	Coverage (%)	Worst-unit unmet	Relocations	Re-plans
0.05	0.00	0.995	2.73	84.0	53,618	181	96
0.05	0.05	0.995	2.73	84.0	53,618	181	43
0.05	0.10	0.995	2.73	84.0	53,618	181	41
0.05	0.20	0.995	2.73	84.0	53,618	181	41
0.05	0.30	0.995	2.73	84.0	53,618	181	41
0.10	0.00	0.990	2.34	76.5	72,972	109	96
0.10	0.05	0.990	2.34	76.5	72,937	109	33
0.10	0.10	0.990	2.34	76.5	72,937	109	20
0.10	0.20	0.990	2.34	76.5	72,937	109	14
0.10	0.30	0.990	2.34	76.5	72,937	109	14
0.20	0.00	0.906	1.80	59.8	109,592	59	96
0.20	0.05	0.906	1.80	59.9	109,592	57	11
0.20	0.10	0.906	1.80	59.9	109,592	55	8
0.20	0.20	0.906	1.80	59.9	109,592	55	6
0.20	0.30	0.906	1.80	59.9	109,592	55	6
0.30	0.00	0.779	1.46	60.1	109,592	64	96
0.30	0.05	0.779	1.46	60.1	109,592	63	27
0.30	0.10	0.779	1.46	59.4	109,592	60	10
0.30	0.20	0.779	1.46	59.4	109,592	60	7
0.30	0.30	0.779	1.46	59.4	109,592	60	7

S3 Sensitivity to α and η, 2019 (CGRR, median over 200 runs)

α	η	Set coverage	Mean set size	Coverage (%)	Worst-unit unmet	Relocations	Re-plans
0.05	0.00	0.982	2.91	30.1	114,358	262	84
0.05	0.05	0.982	2.91	30.2	114,358	260	81
0.05	0.10	0.982	2.91	30.2	114,448	242	66
0.05	0.20	0.982	2.91	31.1	114,660	154	39
0.05	0.30	0.982	2.91	31.4	115,351	128	34
0.10	0.00	0.975	2.42	28.1	119,375	466	84
0.10	0.05	0.975	2.42	28.2	119,305	464	82
0.10	0.10	0.975	2.42	28.1	117,669	391	69
0.10	0.20	0.975	2.42	28.8	117,304	277	55
0.10	0.30	0.975	2.42	28.2	119,345	258	50
0.20	0.00	0.836	1.79	26.4	130,172	381	84
0.20	0.05	0.836	1.79	26.4	130,292	369	81
0.20	0.10	0.836	1.79	26.4	128,932	301	60

0.20	0.20	0.836	1.79	25.6	131,992	125	24
0.20	0.30	0.836	1.79	25.2	128,519	61	13
0.30	0.00	0.708	1.48	27.5	125,648	477	84
0.30	0.05	0.708	1.48	27.5	125,553	467	80
0.30	0.10	0.708	1.48	27.6	123,787	388	62
0.30	0.20	0.708	1.48	26.1	129,422	174	33
0.30	0.30	0.708	1.48	25.7	129,952	109	23

S4 Coverage (%) under alternative mean need shares, 2018

Method	1%	2%	5%
B1	55.3	36.3	17.0
B2	79.6	63.0	36.4
B3	98.9	85.2	54.8
B4	85.7	65.6	35.4
B5	76.3	62.7	43.3
CGRR	93.9	76.5	46.0
FC	78.8	57.7	29.7
FU	75.6	55.6	27.8

S4 Coverage (%) under alternative mean need shares, 2019

Method	1%	2%	5%
B1	47.7	29.5	13.1
B2	45.2	27.8	12.3
B3	43.7	27.6	12.6
B4	43.6	28.2	12.9
B5	45.2	27.8	12.3
CGRR	44.9	28.1	12.4
FC	44.5	28.0	12.6
FU	43.5	27.4	12.4

S5 Run log


```
2026-11-06 16:06:06 START smoke_test=False python=3.13.18 numpy=2.4.6 pandas=2.2.3
2026-11-06 16:06:06 CONFIG {"smoke_test": false, "events": ["2018", "2019"], "data_dir": "/content/drive/MyDrive/Paper4_Kerala", "results_dir":
"/content/drive/MyDrive/Paper4_Kerala/results", "model": {"alpha": 0.1, "eta": 0.1, "beta": 0.5, "tau_max": 12.0, "lam": 0.01, "mu": 0.05, "rho": 0.5, "fixed_width": 0.3,
"b4_gamma_frac": 0.5, "need_beta": [2.0, 98.0], "resources": {"team": {"kappa": 150, "hmax": 0.3, "mode": "road"}, "boat": {"kappa": 60, "hmax": 99.0, "mode": "boat"}, "medical":
{"kappa": 80, "hmax": 0.3, "mode": "road"}}}, "experiment": {"n_runs": 200, "n_scen_b3": 30, "b5_search": 60, "seed": 42, "sensitivity_alpha": [0.05, 0.1, 0.2, 0.3],
"sensitivity_eta": [0.0, 0.05, 0.1, 0.2, 0.3], "demand_means": [0.01, 0.02, 0.05]}}
2026-11-06 16:06:06 [2018] temperature T=1.425  qhat=0.875  delta=[7.00e-06 7.60e-05 2.63e-04 4.61e-04]  Gamma=6
```

```
2026-11-06 16:06:06 [2018] test ECE raw=0.107 -> scaled=0.150; Brier 0.470 -> 0.486
2026-11-06 16:06:10   B1    replans=  1  coverage=36.3%
2026-11-06 16:06:11   B2    replans= 96  coverage=63.0%
2026-11-06 16:06:37   B3    replans= 96  coverage=85.2%
2026-11-06 16:06:45   B4    replans= 96  coverage=65.6%
2026-11-06 16:06:45   B5    replans= 96  coverage=62.7%
2026-11-06 16:06:47   CGRR  replans= 20  coverage=76.5%
2026-11-06 16:06:50   A1    replans= 21  coverage=65.4%
2026-11-06 16:06:55   A2a   replans= 96  coverage=76.5%
2026-11-06 16:06:55   A2b   replans=  1  coverage=56.9%
2026-11-06 16:06:59   A3    replans= 21  coverage=76.6%
2026-11-06 16:07:05   FU    replans= 96  coverage=55.6%
2026-11-06 16:07:09   FC    replans= 96  coverage=57.7%
2026-11-06 16:43:14   S1: best baseline B3
2026-11-06 16:43:14   S2: best baseline B3
2026-11-06 16:43:14   S3: best baseline B3
2026-11-06 16:43:14   S4: best baseline B3
2026-11-06 16:43:37   alpha=0.05 done
2026-11-06 16:43:49   alpha=0.1 done
2026-11-06 16:43:58   alpha=0.2 done
2026-11-06 16:44:09   alpha=0.3 done
2026-11-06 16:44:10 [2018] done
2026-11-06 16:44:10 [2019] temperature T=1.425  qhat=0.875  delta=[7.00e-06 7.60e-05 2.63e-04 4.61e-04]  Gamma=6
2026-11-06 16:44:10 [2019] test ECE raw=0.169 -> scaled=0.151; Brier 0.560 -> 0.556
2026-11-06 16:44:12   B1    replans=  1  coverage=29.5%
2026-11-06 16:44:12   B2    replans= 84  coverage=27.8%
2026-11-06 16:53:40   B3    replans= 84  coverage=27.6%
2026-11-06 17:19:47   B4    replans= 84  coverage=28.2%
2026-11-06 17:19:47   B5    replans= 84  coverage=27.8%
2026-11-06 17:21:23   CGRR  replans= 69  coverage=28.1%
2026-11-06 17:44:01   A1    replans= 68  coverage=28.5%
2026-11-06 17:45:43   A2a   replans= 84  coverage=28.1%
2026-11-06 17:45:44   A2b   replans=  1  coverage=29.5%
2026-11-06 17:45:48   A3    replans= 64  coverage=30.8%
2026-11-06 17:46:05   FU    replans= 84  coverage=27.4%
2026-11-06 17:48:44   FC    replans= 84  coverage=28.0%
2026-11-06 18:05:02   S1: best baseline B1
2026-11-06 18:05:02   S2: best baseline B1
2026-11-06 18:05:02   S3: best baseline B1
2026-11-06 18:05:02   S4: best baseline B1
2026-11-06 18:06:22   alpha=0.05 done
2026-11-06 18:13:20   alpha=0.1 done
2026-11-06 18:19:52   alpha=0.2 done
2026-11-06 18:25:06   alpha=0.3 done
2026-11-06 18:25:07 [2019] done
2026-11-06 18:25:07 END
INPUT_SHA256_16 {"2018": {"forecasts": "1dec4a31025bb1cb", "stocks": "610e0187999926bd", "route_times": "3c0c78b75c9cb3c4", "exposure": "1141856b81f400e1", "tau0":
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"1141856b81f400e1", "tau0": "2be6eec7bf6fe711", "calibration": "49a98d2fdc626456"}}
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